

Speak Like A Mathematician:

concurrent:

point of concurrency:

circumcenter:

incenter:

Circumcenter and Incenter

Circumcenter Theorem: the circumcenter of a triangle is equidistant from the _____ of the triangle.

Incenter Theorem: the incenter of a triangle is equidistant from the _____ of the triangle.

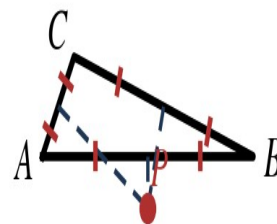
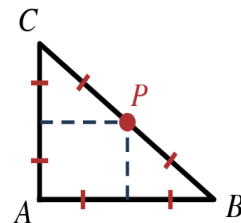
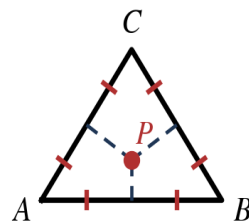
Where the circumcenter lands: acute $\rightarrow$ _____ the triangle $\cdot$ right $\rightarrow$ _____ the triangle $\cdot$ obtuse $\rightarrow$ _____ the triangle.

The incenter is _____ **inside the triangle, no matter what kind of triangle it is.**

Example 1 - Where Is the Circumcenter?

Each triangle at the right shows its three perpendicular bisectors meeting at P.

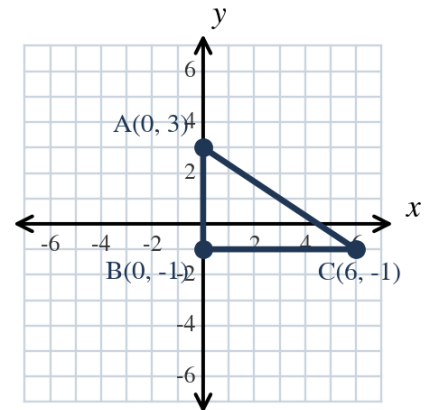
- Describe where P lands in an acute triangle.
- Describe where P lands in a right triangle.
- Describe where P lands in an obtuse triangle.
- Why is $PA = PB = PC$ in every case?



Example 2 - Find a Circumcenter in the Coordinate Plane

Find the coordinates of the circumcenter of $\triangle ABC$ with $A(0, 3)$, $B(0, -1)$, and $C(6, -1)$.

- $\overline{AB}$ is vertical. Write an equation of its perpendicular bisector.
- $\overline{BC}$ is horizontal. Write an equation of its perpendicular bisector.
- Where do the two perpendicular bisectors meet?
- Check with the Circumcenter Theorem: find the distance from that point to each vertex.



Example 3 - A Second Circumcenter

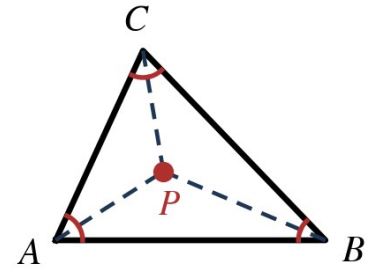
Find the circumcenter of $\triangle DEF$ with $D(-2, 4)$, $E(-2, -2)$, and $F(6, -2)$.

- Write the two easy perpendicular bisectors.
- Give the circumcenter and the common distance.

Example 4 - The Incenter

In the diagram, P is the incenter of the triangle and the marked segments are perpendicular to the sides.

- What does the Incenter Theorem tell you about those three segments?
- If one distance is $3x + 5$ and another is $5x - 7$, find x and the common distance.
- Why is the incenter always inside the triangle?



Example 5 - Which Center Do You Want?

Decide whether each situation calls for the circumcenter or the incenter. Explain.

- A cell tower must be the same distance from three towns.
- A sprinkler must be the same distance from the three fences bounding a triangular yard.
- You want the largest circle that fits inside a triangular garden.
- You want the circle that passes through all three corners of a triangular table.